

National CENTRALIZED HEALTHCARE DATABASE SYSTEM (NCHDS)

An Interoperable Digital Health & Policy Framework for 175 Million Citizens of Bangladesh



Presenter

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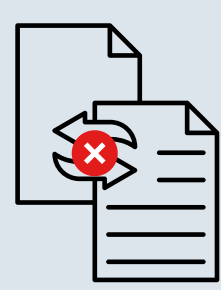
Vision

1 Patient, 1 Lifelong
Health Record.

ABSTRACT

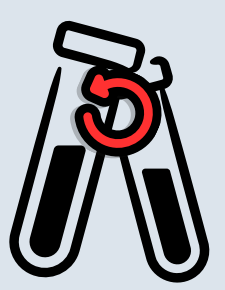
Bangladesh's healthcare delivery is severely constrained by paper-centric, fragmented records and isolated electronic registries. Aligning with the Prime Minister's "Smart Bangladesh 2041" vision to build a digitally integrated, equitable healthcare ecosystem, this research proposes the National Centralized Healthcare Database System (NCHDS). Designed to secure and centralize the health records of 175 million citizens, NCHDS links patient histories to a **unique National Health ID (NHID)** derived from NID/BRIS registries. The enterprise architecture leverages a microservices framework with HL7 FHIR messaging, Apache Kafka event pipelines, and sharded PostgreSQL clusters to comfortably process 50,000+ requests per second. By enabling longitudinal care, eliminating diagnostic redundancies, and tracking counterfeit pharmaceuticals in real time, NCHDS is projected to save over \$90M annually. This system provides a transformative blueprint for universal digital health coverage.

PROBLEM STATEMENTS



Fragmented Records

Patient records do not transfer between public tertiary medical complexes resulting in complete diagnostic blindness.



Diagnostic Redundancy

Repeated & complex diagnostic tests waste time, money and resource as past records are inaccessible.



Poor Continuity

Unauthorized medicine sales operate undetected without real-time patient prescription limits & BIN/TIN verification.



Limited Oversight

Hospital reporting & audit logs are paper-centric, leading to, administrative friction, & high operational overhead.

RESEARCH FINDINGS & PROJECTIONS

30%+

Reduction in out of pocket health expenditure

\$90M

Annual Saving via reduced duplication & elimination. NPV of \$114.5M in 5Yrs

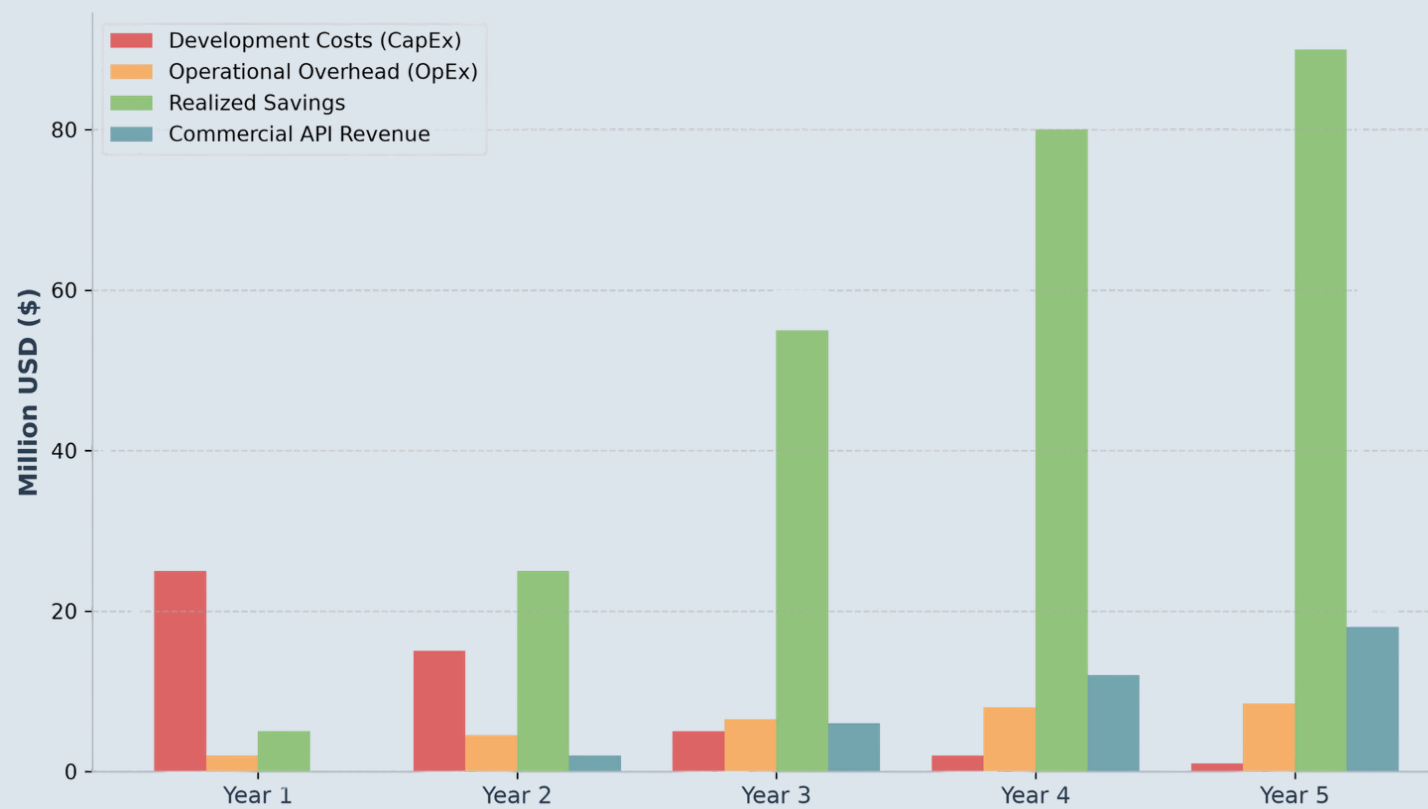
Over 50,000

Write operations /sec to millions of daily clinical hits.

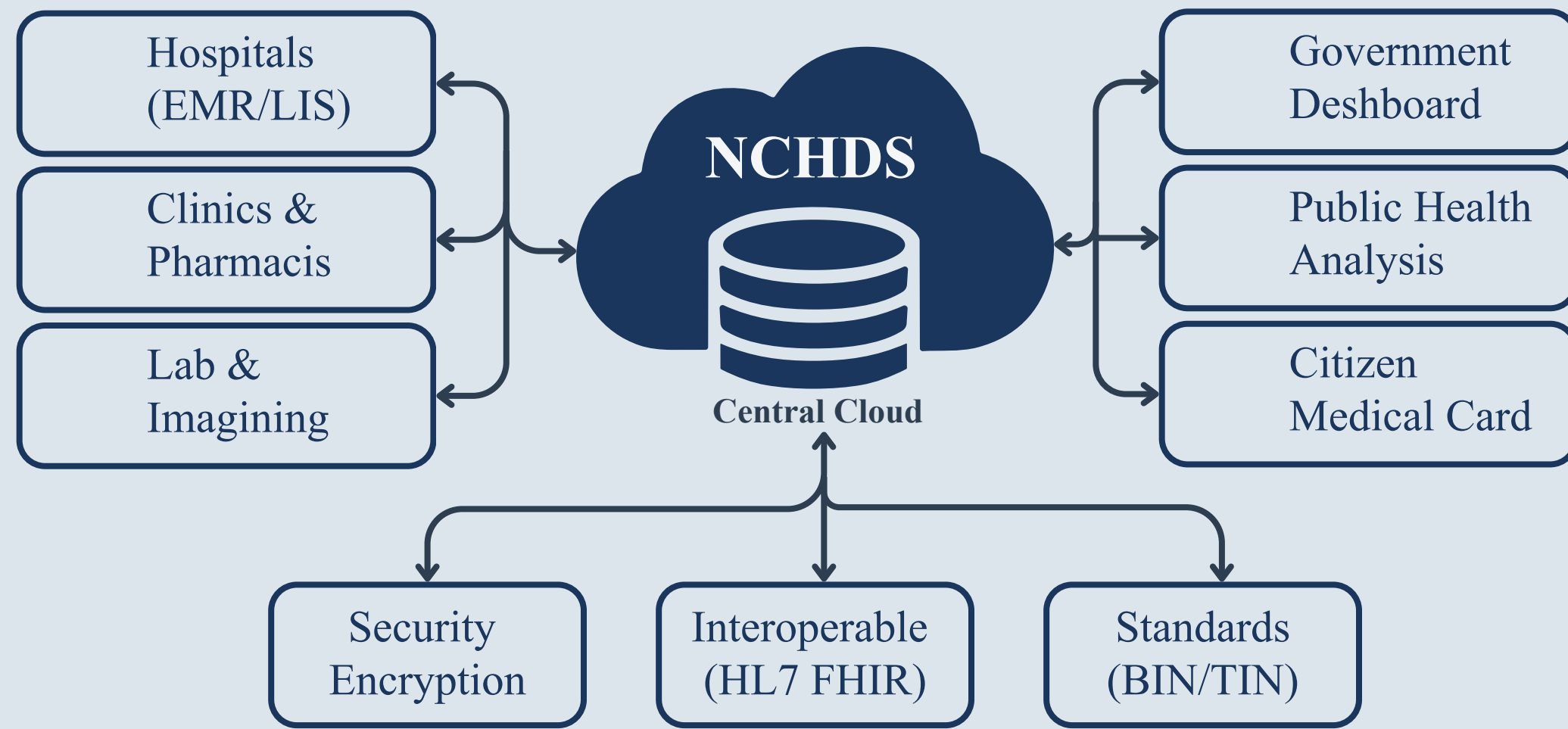
175M

Citizen with one lifelong digital health record

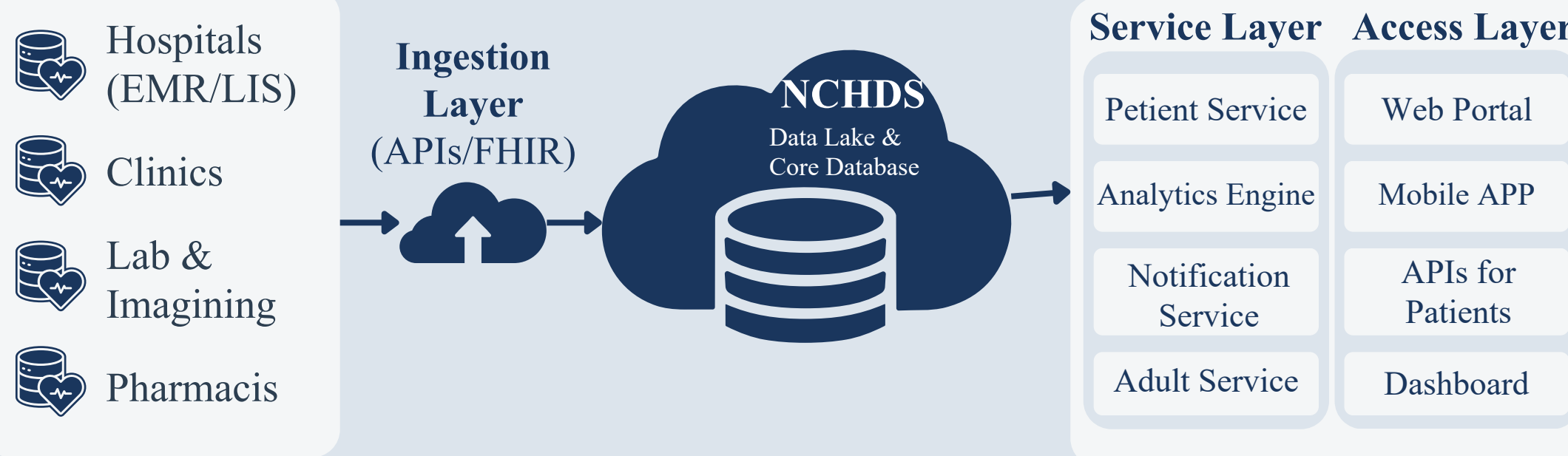
NCHDS 5-Year Financial Projections (Costs vs. Economic Returns)



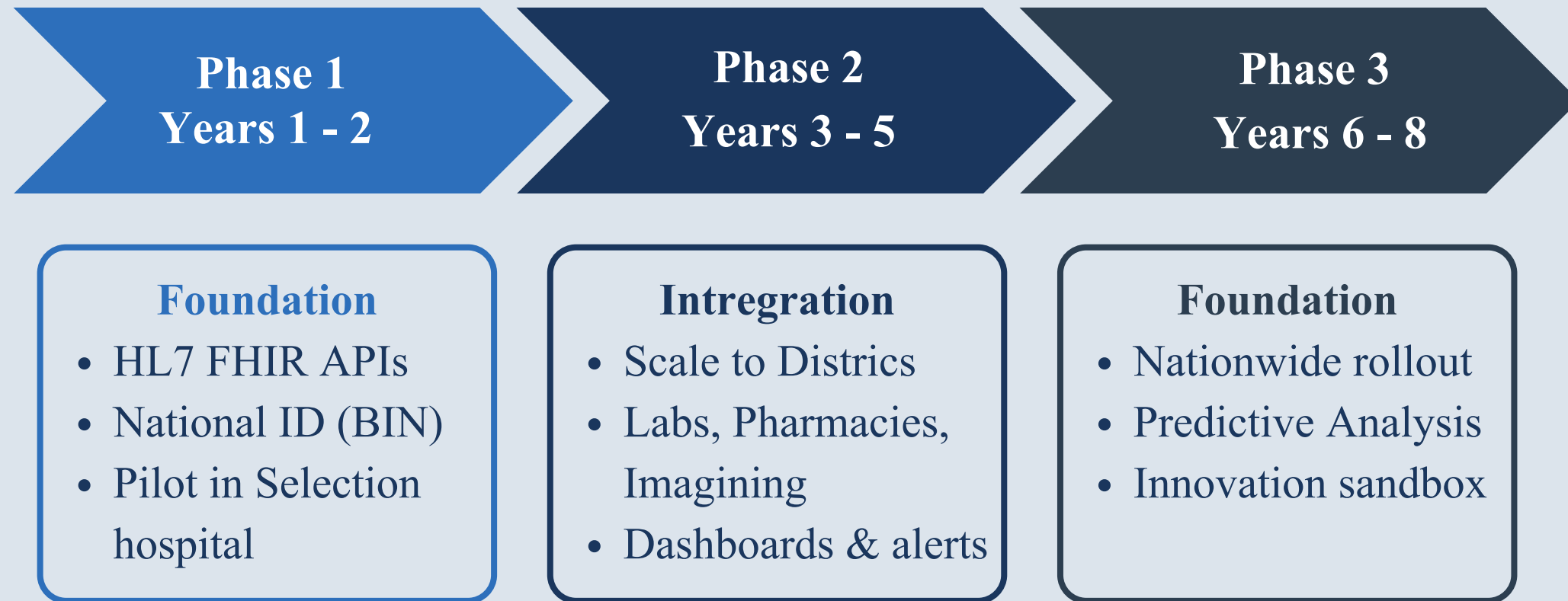
OUR SOLUTION: NCHDS



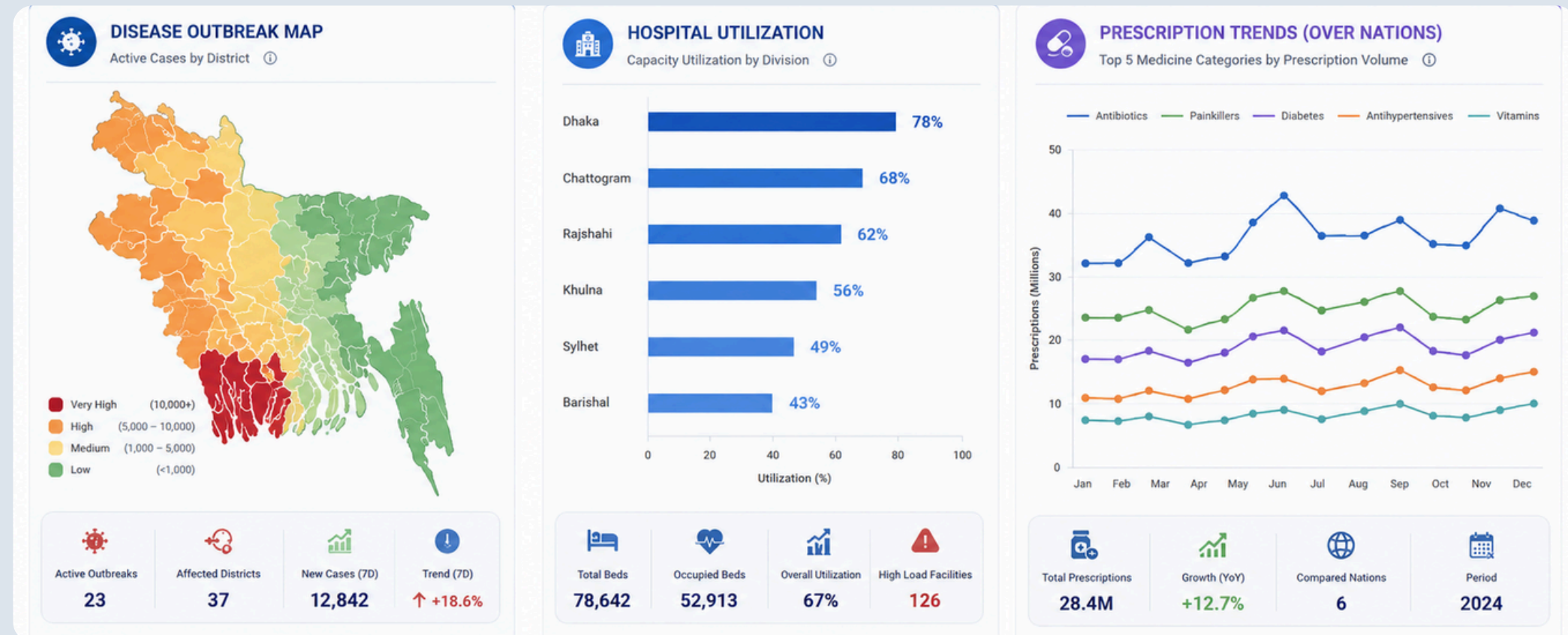
TECHNICAL ARCHITECTURE



IMPLEMENTATION ROADMAP (8 Year Plan)



NATIONAL DASHBOARD & INSIGHTS



MEDICAL CARD: "ONE CARD-LIFELONG RECORD"

NATIONAL DIGITAL HEALTH CARD
One Patient, One Lifelong Health Record.

HEALTH SUMMARY

- Allergies: Penicillin, Pollen
- Chronic Conditions: Hypertension
- Current Medications: Amlodipine 5mg (Daily)
- Immunizations: Up to date
- Last Check-up: 12-Apr-2025
- Primary Care Provider: Dr. A. Rahman, Dhaka Medical College Hospital

EMERGENCY CONTACT

- Name: Shamima Akter
- Relation: Wife
- Mobile: 01XXXXXXX

Issued: 01-Jan-2025

Secure • Private • Interoperable

Valid Till: 01-Jan-2035

What's Inside?

- ✓ Health Summery
- ✓ Visit & Treatment History
- ✓ Lab & Image Reports
- ✓ Medication & Prescriptions
- ✓ Immunization Records
- ✓ Insurance Details
- ✓ Emergency Contacts
- ✓ Care Tean & Provider

KEY FEATURES



Interoperable
(HL7 FHIR)



Secure &
Compliant



Real-time
Monitoring



Scalable
Cloude Native



Evidance-based
Policy Support

WHY NOW?

- 🏥 Rising healthcare concern, demand & costs
- 🌐 Pandamc exposed data fragmentation
- 📍 Digital Bangladesh Momentum
- 📋 Policy alignment with smart Bangladesh 2041

REFERENCES

- [1] M. Sarker et al., "Synergies and fragmentations of universal health coverage in Bangladesh," Lancet Reg. Health Southeast Asia, vol. 7, 2022.
- [2] M. Bouh et al., "Impact of limited access to digital health records on doctors," Digital Health, vol. 10, 2024.
- [3] M. A. H. Khan, "Bangladesh's digital health journey: reflections on a decade of quiet revolution," WHO SEAJPH, vol. 8, 2019.



Scan to
see demo
of
NCHDS
Prototype

CONCLUSION

NCHDS unifies Bangladesh's health data ecosystem to deliver better care, reduce waste, enable early outbrake detection, and support data driven policy. TOgether, we move toward a healthier nation

"One Patient, One Lifelong Health Record."